

Isobaric vapor–liquid phase equilibria for the binary systems of *tert*-butanol + 2-ethyl-1-hexanol and *n*-butanol + 2-ethyl-1-hexanol

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Abstract

Isobaric vapor–liquid equilibria for the binary mixtures of *tert*-butanol (TBA) + 2-ethyl-1-hexanol and *n*-butanol (NBA) + 2-ethyl-1-hexanol were experimentally investigated at atmospheric pressure in the temperature range of 353.2–458.2 K. The raw experimental data were correlated using the UNIQUAC and NRTL models and used to estimate the interaction parameters between each pair of components in the systems. The experimental activity coefficients were obtained using the gas chromatographic method and compared with the calculated data obtained from these equilibrium models. The results show that UNIQUAC model gives better correlation than NRTL for these binary systems. The liquid–liquid extraction of TBA from aqueous solution using 2-ethyl-1-hexanol was demonstrated by simulation and the variation of separation factor of TBA at several temperatures was reported.

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1. Introduction

The precise vapor–liquid equilibrium (VLE) data of binary mixtures like alcohol–alcohol are important to design many chemical processes and separation operations. The VLE investigations of binary systems have been the subject of much interest in recent years [1–9]. In the past, several authors have reported isobaric VLE data for the system of *tert*-butanol + water [10–13]. Darwish and Al-Anber [14] have presented isobaric VLE data for the binary systems of *tert*-butanol + water and of *tert*-butanol + isobutanol at 94.9 kPa.

We have recently reported the LLE data for ternary mixture of (water + TBA + 2-ethyl-1-hexanol) [15] and (water + *n*-butanol + 2-ethyl-1-hexanol) [16]. From the experimental results, 2-ethyl-1-hexanol was chosen as a very good solvent for recovering TBA and NBA from aqueous solution [17]. This solvent has been also used as an organic solvent to

determination of LLE data for some ternary mixtures of (water + ethanol + 2-ethyl-1-hexanol) [18] (water + acetone + 2-ethyl-1-hexanol) [19], and (water + acetic acid + 2-ethyl-1-hexanol) [20].

In order to know the activity coefficients of the binary systems of TBA + 2-ethyl-1-hexanol and NBA + 2-ethyl-1-hexanol, the VLE data are also needed. These experimental data are determined using a method based on gas chromatographic technique described previously [17]. The experimental activity coefficients and VLE data can be correlated to several equilibrium methods such as UNIQUAC and NRTL models. Equilibrium models, such as the universal quasi-chemical model (UNIQUAC) [21] and the non-random, two-liquid model (NRTL) [22] have been successfully applied for the correlation of several liquid–liquid and vapor–liquid systems. These models depend on optimized interaction parameters between each pair of components in the systems, which can be obtained by experiments.

In this research we have presented experimental activity coefficients and isobaric VLE data for the binary systems TBA + 2-ethyl-1-hexanol and NBA + 2-ethyl-1-hexanol

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at 101.3 kPa. The calculated activity coefficients have been correlated by fitting the binary interaction parameters of different models UNIQUAC and NRTL.

2. Experimental

2.1. Materials

tert-Butanol, *n*-butanol and 2-ethyl-1-hexanol were obtained from Merck with purity better than 99% and were used without further purification. The purity of these materials was checked by gas chromatography.

2.2. Apparatus and procedure

The VLE measurements were carried out using a Labodest unit built by Fischer, and equipped with a Cottrell pump. The apparatus and procedure have been described completely by Artigas [1] and Walas [23]. The temperature of the cell was maintained with an accuracy of within ± 0.1 K. The composition analysis of the vapor and liquid phase were determined using Konik gas chromatography (GC) equipped with a thermal conductivity detector (TCD) and Shimadzu C-R2AX integrator. The column, injector and detector temperatures were 493.2, 404.2 and 433.2 K, respectively. A 2 m $\times$ 2 mm column was used to separate the components.

3. Results and discussion

The experimental and predicted activity coefficients of the components, γ_i , are compiled in Tables 1 and 2. The same data for the binary systems are graphically represented in Figs. 1 and 2.

The experimental activity coefficients of the components, γ_i , were obtained using the gas chromatographic (GC) method [17] and by means of the following equation:

$$A_i = K_i n_i \quad (1)$$

where A_i is the pick area and n_i the numbers of moles of the component i . K_i is a constant of proportionality. For each species i , the numbers of moles can be related to the fugacity of component i in the vapor phase, f_i^v , using the following equation:

$$f_i^v = RT \frac{n_i}{V} \exp \left[\frac{2}{v} \sum_j^m y_j B_{ij} \right] \quad (2)$$

where V is the volume of the sample, n_i the numbers of moles of the component i , B_{ij} the second virial coefficient. For a liquid mixture in equilibrium with a vapor mixture at the same temperature and pressure, the condition of thermodynamic equilibrium is given by

$$f_i^l = f_i^v \quad (3)$$

Table 1
Experimental and predicted activity coefficient for the binary system of TBA + 2-ethyl-1-hexanol

Activity coefficient TBA			Activity coefficient 2-ethyl-1-hexanol		
Experimental	UNIQUAC	NRTL	Experimental	UNIQUAC	NRTL
4.7600	4.3899	4.2981	1.0000	1.0000	1.0000
2.2834	3.3237	2.3596	1.1700	1.0250	1.1070
2.1000	2.5914	2.0012	1.1900	1.0500	1.1120
2.3990	1.7175	1.9945	1.3000	1.2538	1.2290
1.2698	1.2759	1.1980	2.0600	1.8976	1.3252
1.2800	1.0639	1.1901	2.7300	2.6420	1.8445
1.1000	1.0158	1.0789	3.2800	3.1053	3.2970
1.0000	1.0001	1.0003	4.6700	3.9214	4.4600
RMSD%	3.14	4.81		6.28	6.43

Table 2
Experimental and predicted activity coefficient for the binary system NBA + 2-ethyl-1-hexanol

Activity coefficient NBA			Activity coefficient 2-ethyl-1-hexanol		
Experimental	UNIQUAC	NRTL	Experimental	UNIQUAC	NRTL
4.5500	4.5182	4.9910	1.0000	1.0000	1.0000
3.5700	3.3107	3.5758	1.1500	1.0164	1.0175
2.8200	2.5293	2.6902	1.2499	1.0657	1.0698
1.7720	1.7547	1.7248	1.4101	1.2772	1.2931
1.3500	1.2416	1.2729	1.6900	1.7005	1.7511
1.1220	1.0537	1.0630	2.8200	2.4901	2.6672
1.0000	1.0129	1.0156	3.5201	3.1131	3.4555
1.0000	1.0000	1.0000	4.3000	3.9704	4.6419
RMSD%	5.74	10.28		9.82	8.92

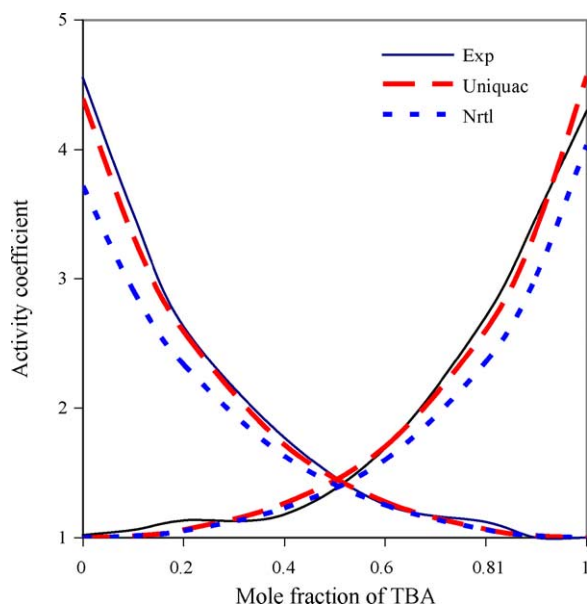


Fig. 1. Activity coefficients in the system of TBA + 2-ethyl-1-hexanol.

where f_i^l and f_i^v are liquid and vapor fugacities. If the liquid phase is a pure component, the fugacity of pure liquid f_i^0 is given by

$$f_i^0 = f_i^v \quad (4)$$

Activity coefficient γ_i is related to x_i and to f standard-state fugacity f_i^0 :

$$\gamma_i = \frac{a_i}{x_i} = \frac{f_i^l}{x_i f_i^0} \quad (5)$$

where a_i is the activity of component i . The activity coefficients can be expressed in function of the areas measured in

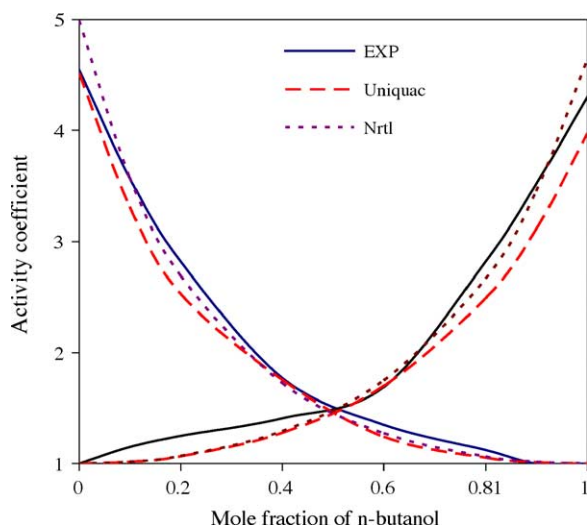


Fig. 2. Activity coefficients in the system of NBA + 2-ethyl-1-hexanol.

Table 3

Vapor–liquid equilibria for *tert*-butanol (TBA) with 2-ethyl-1-hexanol

T (K)	X_1	Y_1	T (K)	X_1	Y_1
459.7300	0.0000	0.0008	368.9010	0.3000	0.9730
445.5325	0.0050	0.2984	366.7212	0.4399	0.9797
436.1964	0.0100	0.4755	363.8160	0.5100	0.9819
428.8066	0.0150	0.5892	361.6598	0.6499	0.9858
422.7940	0.0199	0.6667	360.6300	0.7200	0.9877
388.2300	0.0899	0.9200	359.5657	0.7899	0.9899
380.5000	0.1599	0.9542	357.0900	0.8599	0.9925
372.1293	0.2299	0.9666	355.5700	0.9300	0.9957

the chromatogram, in the following form:

$$\gamma_i = \frac{A_i}{x_i A'_i} \left\{ \frac{\exp \left[\left(\frac{2}{V} \right) \sum_j^m n_j B_{ij} \right]}{\exp \left[\left(\frac{2}{V} \right) \sum_j^{m'} n_j B_{ij} \right]} \right\} \quad (6)$$

where V is the volume of the sample, n_i and n_j are the numbers of moles of the component i and j , respectively, B_{ij} the second virial coefficient, A' the standard area for pure component and A_i the area of the pick of component i . As the exponential terms of Eq. (1) approximately spread to the unit, Eq. (6) reduces to

$$\gamma_i = \frac{A_i}{x_i A'_i} \quad (7)$$

The UNIQUAC and NRTL models were used to correlate the experimental activity coefficients and VLE data. As it can be seen from Figs. 1 and 2, the experimental data (solid line) are in good agreement with the predicted data (dashed line). Although the both equations adequately fit the experimental data for these binary systems, however, the UNIQUAC model fit better the experimental data. The non-ideal behavior of the both systems is quiet obvious in these diagrams. A positive deviation is observed for the systems as indicated by the positive values of activity coefficients over the composition range of 0.1–0.9.

The experimental isobaric VLE data (T , x , y) for the binary systems (*tert*-butanol + 2-ethyl-1-hexanol) and (*n*-butanol + 2-ethyl-1-hexanol) studied in this work are listed in Tables 3 and 4. The same experimental data together with calculated VLE data are graphically shown in Figs. 3 and 4. The dashed line in theses diagrams were obtained from the UNIQUAC and NRTL equations. Antoine coefficients were

Table 4

Vapor–liquid equilibria for *n*-butanol (NBA) with 2-ethyl-1-hexanol

T (K)	X_1	Y_1	T (K)	X_1	Y_1
458.15	0.0000	0.0008	402.8900	0.3000	0.8703
452.6213	0.0049	0.2994	400.9500	0.3400	0.8845
444.5697	0.0112	0.3245	399.5000	0.4400	0.9110
441.3355	0.0200	0.3980	398.2821	0.5100	0.9234
418.6233	0.0900	0.7289	396.1976	0.6500	0.9235
409.6233	0.1685	0.8125	393.3350	0.8600	0.9567
405.6594	0.2300	0.9480	392.0000	0.9270	0.9768

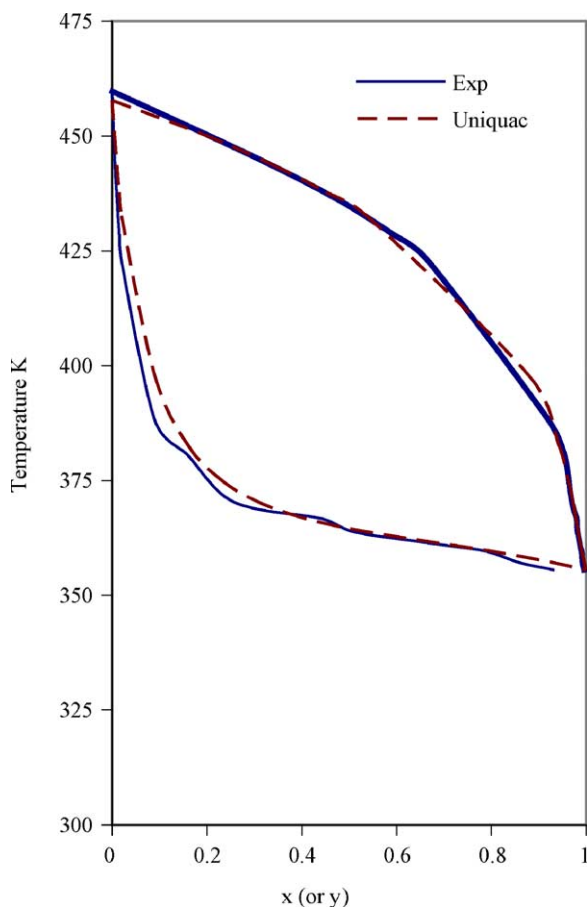


Fig. 3. Boiling temperature diagram (T) for the system of TBA + 2-ethyl-1-hexanol.

used in calculating vapor pressures of *tert*-butanol, 2-ethyl-1-hexanol, and *n*-butanol. Antoine coefficients (Table 5) were obtained by applying the equation:

$$\ln p_i^0 = A - \frac{B}{T - C} \quad (8)$$

Vapor pressures, p_i^0 (mmHg), were fitted with the Antoine equation. In these T - x - y diagrams, it can be seen that the binary systems do not show azeotropic behavior as the boiling point of 2-ethyl-1-hexanol is quite far from that of NBA and TBA. In contrast, the binary systems of water–TBA and water–NBA show azeotropic behavior.

In order to indicate the ability of the solvent in the recovery of TBA/NBA, the separation factor (S) was measured. Fig. 5 shows the variation of separation factor of TBA from aqueous solutions using 2-ethyl-1-hexanol as a function of the mole

Table 5
Antoine coefficients

Component	A	B	C
<i>n</i> -Butanol	17.2160	3137.02	−94.43
<i>tert</i> -Butanol	16.8548	2658.29	−95.50
2-Ethyl-1-hexanol	15.3614	2773.46	−140.00

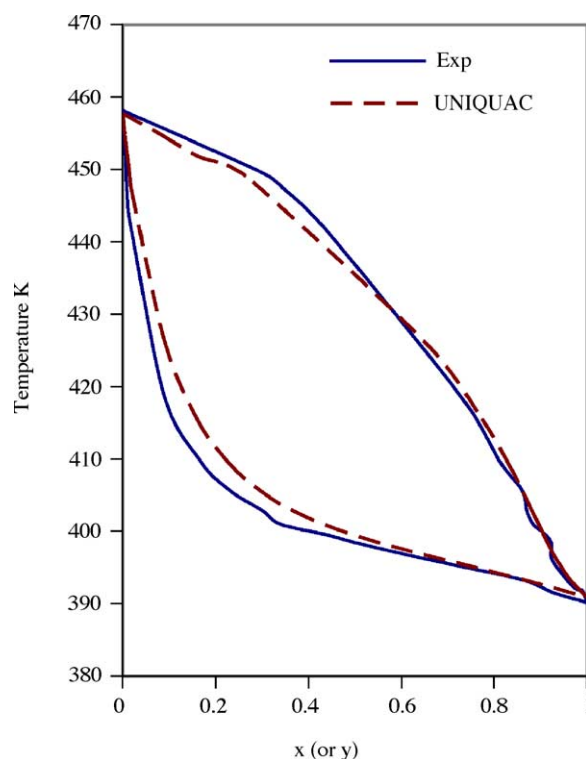


Fig. 4. Boiling temperature diagram (T) for the system of NBA + 2-ethyl-1-hexanol.

fraction of TBA in organic phase at several temperatures. It can be seen that the separation factor increases sharply at 315.2 K and below the 0.38 mol fraction of TBA in organic phase, indicating an optimum condition of temperature and concentration for the separation.

The experimental results indicated that 2-ethyl-1-hexanol has high separation factor, which shows the ability of this solvent to extract TBA or NBA from water. So, due to the several advantages such as high separation factor, high boiling point,

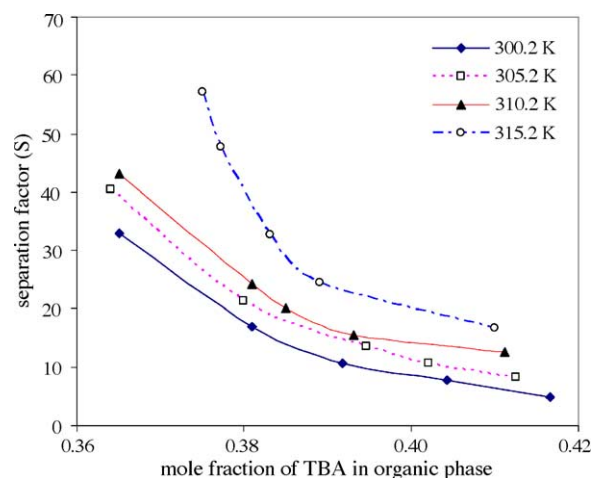


Fig. 5. Separation factor S of TBA as a function of the mole fraction of TBA in organic phase at different temperatures.

Table 6
UNIQUAC and NRTL parameters the systems TBA + 2-ethyl-1-hexanol and NBA + 2-ethyl-1-hexanol

<i>i</i>	<i>j</i>	UNIQUAC		NRTL	
		U_{12} (K)	U_{21} (K)	Δg_{12}	Δg_{21}
TBA	2-Ethyl-1-hexanol	−55.4544	215.7780	511.769	480.0741
NBA	2-Ethyl-1-hexanol	−78.3030	247.1739	384.4047	578.2420

Table 7
The UNIQUAC structural parameters

Components	<i>R</i>	<i>q</i>
TBA	3.45	3.05
NBA	3.45	3.05
2-Ethyl-1-hexanol	6.15	5.02

low cost and very low solubility in water, this solvent can be considered as a good solvent for recovering TBA and NBA from the aqueous solution.

Table 6 shows the calculated value of the UNIQUAC and NRTL binary interaction parameters for the mixture TBA + 2-ethyl-1-hexanol and NBA + 2-ethyl-1-hexanol using universal values for the UNIQUAC and NRTL parameters. The mixture non-randomness parameter α_{12} in the NRTL equation was fixed at 0.3. The values of *r* and *q* used in the UNIQUAC equation are presented in Table 7. The UNIQUAC structural parameters *r* and *q* were calculated from group contribution data that has been previously reported [22]. The UNIQUAC and NRTL equations were optimized by minimizing the following objective function (OF):

$$OF = \sum_{j=1}^{np} \sum_{i=1}^{nc} \left[\frac{\gamma_{ij}^{\text{exp}} - \gamma_{ij}^{\text{cal}}}{\gamma_{ij}^{\text{exp}}} \right]^2 \quad (9)$$

where γ_{ij} are the corresponding activity coefficients and *np* is the number of experimental data and *nc* is number components.

The goodness of fit, between the observed and calculated mole fractions, was calculated in terms of the root mean square deviation (RMSD). The RMSD values were calculated according to the equation of percentage root mean square deviation. Values of RMSD given by these two models are listed in Table 8. From these values it can be concluded that

Table 8
RMSD% values for the UNIQUAC and NRTL

	NBA	TBA
UNIQUAC	5.74	3.14
	9.82	6.28
Average	7.78	4.71
NRTL	10.28	4.81
	8.92	8.05
Average	9.6	6.43

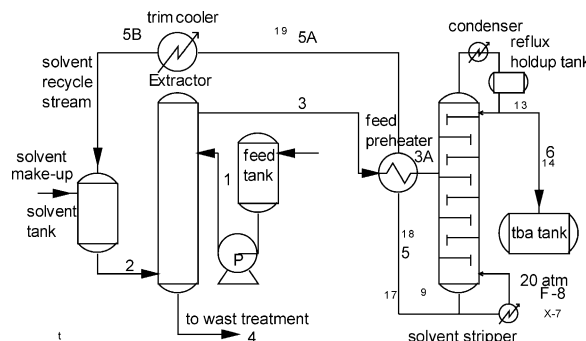


Fig. 6. TBA process separation flow diagram using extraction with 2-ethyl-1-hexanol.

the UNIQUAC model gives better prediction than NRTL for these binary systems.

3.1. Simulation and separation process

A commercial simulation program, ChemCad III, was used for simulation of the fractional distillation column. The flow diagram for TBA purification process is shown in Fig. 6. The product (TBA) from the catalytic hydration of isobutene to *tert*-butanol on Amberlyst-15 in a co-current down flow trickle-bed reactor [24] is sent to an extraction column prior to stripper. The TBA concentrations used in our design are 20 and 80 wt.% water [24]. By removing water from the product flow, the TBA concentration the top of the distillation column is 99.5 wt.%. In this purification process, the production of TBA is estimated to be about 30 t per day [25]. The stripper column was optimized at 23 plates with feed entering at plate 14. The total height of this column is 9 m and stripper diameter is 3 m. The values for optimization were estimated from

Table 9
Detailed materials and stream compositions (currents of the process)

Stream ID	1	2	3	4	3A	5	5A	5B	6
TBA	0.2	—	0.143	0.001	0.143	0.001	0.001	0.001	0.999
Water	0.8	—	0.001	0.999	0.001	0.000	0.000	0.000	0.000
2EH	0.0	1.0	0.856	0.000	0.856	0.999	0.999	0.999	0.001
Totals (kg/h)	66603.2	74000	94820.4	45782.8	94820.5	81890.6	81890.6	81890.6	12930
Temperature (K)	298.2	298.2	298.2	298.2	423.2	455.2	323.2	298.2	298.2
<i>P</i> (atm)	1	1.0	2.7	1.0	2.0	1.0	2.0	1.0	1.0

the method of McCabe [26], and Coulson and Richardson [27] and also described by Dadgar and Foutch [25]. Details of process for the TBA purification set together with operation conditions (T , P) are presented in Table 9. The operation conditions were selected using the VLE data. It should be noted that, in this table concentration is in terms of weight fraction. The simulation indicated recovery of the TBA by extraction to be facile.

4. Conclusion

Isobaric VLE data at 101.3 kPa were measured for two binary mixtures, i.e. TBA + 2-ethyl-1-hexanol and NBA + 2-ethyl-1-hexanol. The UNIQUAC and NRTL models were used to correlate the experimental activity coefficients and VLE data. Deviation from ideal solution behavior was observed for both binary systems, whereas none of them showed azeotropic behavior.

The UNIQUAC model fit the experimental data for the systems of TBA + 2-ethyl-1-hexanol and NBA + 2-ethyl-1-hexanol with an average root mean square deviation (RMSD) of 4.71 and 6.43% respectively. The NRTL model gave higher values for the systems, i.e. 7.78 and 9.59% respectively. From these values it can be concluded that the NRTL model gives poor correlations, whereas the UNIQUAC model adequately fit the experimental VLE data for these binary systems.

List of symbols

a	optimized interaction parameter
A , B , and C	Antoine equation parameters
B_{ij}	second virial coefficient
C	number components
q_i	relative surface area per molecule
r_i	number of segments per molecule
RMSD	root mean square deviation
T	absolute temperature (K)
u_{ij}	interaction energy
x	mole fraction
x_i	equilibrium mole fraction of component i

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